

Emergency Liquidity, Borrowing, and Hardship

A Descriptive SHED 2025 Microdata Note

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Abstract

This paper uses 2025 public-use microdata from the Federal Reserve's Survey of Household Economics and Decisionmaking (SHED) to describe how emergency liquidity, borrowing-style emergency responses, and concrete hardship proxies vary together among U.S. adults. It is a descriptive working paper intended to convert an earlier public-indicator note into respondent-level evidence with transparent denominators, approximate uncertainty, and replication code. In weighted 2025 estimates, 63.1 percent of adults could cover a \$400 emergency expense with cash or its equivalent, 54.8 percent had three-month emergency savings, 15.3 percent would put a \$400 emergency on a credit card and pay over time, and 12.2 percent would not be able to pay the \$400 expense right now. Adults lacking both \$400 cash/equivalent capacity and three-month emergency savings were much more likely to report hardship-related outcomes: 62.9 percent were financially vulnerable, 37.2 percent could not pay the \$400 expense right now, 61.5 percent had bill pressure, 40.1 percent received outside-household help with expenses, and 38.6 percent were credit constrained. Approximate weighted linear-probability models with controls for income, education, age, race/ethnicity, tenure, employment, children, and region preserve large low-buffer associations. The core conclusion is narrow: emergency-buffer variables are not causal estimates and partly overlap with subjective financial self-assessment, but they identify a materially different hardship profile in the SHED microdata.

1. Introduction

The earlier papers in this sequence used public indicators to map household financial stress across housing entry costs, asset exposure, essential costs, labor adaptation, long-horizon formation, fiscal space, credit, expectations, resilience costs, and emergency savings. The capstone synthesis argued that the next useful step should be narrower and more empirical. Instead of adding another broad channel, the research program should test one mechanism with household-level microdata.

This paper takes a first step in that direction. It uses 2025 SHED public-use microdata to study emergency liquidity, borrowing-style emergency responses, and hardship-related outcomes. The paper's question is descriptive:

In the 2025 SHED microdata, how do reported hardship outcomes differ between adults with and without short-run emergency liquidity buffers?

That question is intentionally modest. Adults who say they can handle a \$400 emergency and have three-month savings are expected to report better financial conditions. The paper therefore should not be read as discovering a surprising new causal fact. Its contribution is more practical: it documents the respondent-level magnitude of these gaps, separates cash/equivalent capacity from borrowing over time, adds concrete hardship proxies beyond general financial self-assessment, documents denominators and skip patterns, reports approximate uncertainty, and creates a reproducible Phase A build for deeper multi-year work.

The main descriptive pattern is strong. Adults lacking both \$400 cash/equivalent capacity and three-month emergency savings are much more likely to report financial vulnerability, inability to pay a \$400 expense right now, bill pressure, outside-household help with expenses, credit constraints, and medical debt. These relationships remain large in approximate weighted linear-probability models with standard demographic and household controls. That does not solve identification. It does establish that emergency-buffer measures contain meaningful information about multiple hardship margins in the public-use file.

2. Contribution and relationship to prior research

This draft is better understood as a descriptive household-finance note than as a completed journal article. It sits near four literatures.

First, research on liquidity constraints and hand-to-mouth households shows that wealth and liquidity are not the same. Kaplan, Violante, and Weidner's "wealthy hand-to-mouth" framework is especially relevant because households may have illiquid wealth while remaining exposed to short-run cash-flow shocks. The SHED \$400 and three-month-savings measures speak directly to liquid shock absorption rather than net worth alone.

Second, the financial-fragility literature, including Lusardi, Schneider, and Tufano, treats emergency-expense capacity as a practical threshold. This paper follows that tradition but updates the descriptive evidence using 2025 SHED public-use microdata and adds respondent-level cross-tabs across several hardship proxies.

Third, household-credit research emphasizes that credit can smooth consumption or deepen fragility depending on terms, repayment capacity, and shock persistence. This paper therefore distinguishes credit-card use paid in full from credit-card borrowing over time. A card paid in full may be a payment instrument; carrying a \$400 emergency over time is closer to borrowed shock absorption.

Fourth, the paper touches the measurement literature on subjective financial well-being. The general "doing at least okay financially" measure is useful but conceptually close to perceived emergency capacity. For that reason, the revised draft gives more emphasis to concrete outcomes: inability to pay the \$400 expense right now, bill pressure, outside-household help, credit constraint, medical debt, and unexpected medical expenses.

The paper does not claim novelty at the level expected from a top general-interest economics journal. Its value is as a transparent empirical bridge: from public aggregate indicators toward a multi-year SHED microdata paper with better outcomes, richer controls, and stronger uncertainty treatment.

3. Data

The data are from the Federal Reserve's 2025 Survey of Household Economics and Decisionmaking public-use CSV file. The Phase A build downloads the official public-use ZIP, extracts `public2025.csv`, and selects

variables needed for emergency-buffer analysis. The file contains 12,934 adult respondents and 816 columns in the public-use release.

The analysis uses the respondent weight `weight`, which the SHED codebook describes as the common weight for single-year cross-sectional statistics and the weight used for the Federal Reserve's published 2025 well-being statistics. The codebook also includes `weight_pop`, `panel_weight`, and `panel_weight_pop`. This draft does not use panel weights because it analyzes only the 2025 cross-section.

The unit of observation is the adult respondent. Some questions refer to spouses, partners, households, or household expenses, but the analysis avoids treating every response as a complete household balance sheet.

4. Measures, denominators, and skip patterns

A central revision after external review is explicit denominator documentation. Several SHED variables are available for all respondents, while others are conditional or constructed. Rates should not be interpreted until the base population is clear.

Table A. Variable audit and approximate uncertainty.

Variable	Base / type	N	Missing %	Weighted rate (%)	Approx. 95% CI (%)
pay_casheqv	All respondents; Federal Reserve constructed binary	12,932	0.0	63.1	62.3–64.0
EF1	All respondents; three-month rainy-day savings	12,934	0.0	54.8	53.9–55.7
EF2	Follow-up among those without three-month rainy-day funds	5,607	56.6	33.7	32.4–35.1
EF3_b	All respondents; \$400 card paid over time option	12,934	0.0	15.3	14.6–16.0
EF3_h	All respondents; unable to pay \$400 right now option	12,934	0.0	12.2	11.6–12.8
atleast_okay	All respondents; Federal Reserve constructed binary	12,933	0.0	72.9	72.1–73.7
bill_pressure	Author proxy: did not pay all bills in full or had difficult bills	12,934	0.0	27.8	27.0–28.6
credit_constrained	Author proxy from credit desire/denial/limited approval	12,934	0.0	18.6	17.9–19.4
bnpl_late_or_fee	Conditional among observed BNPL users	2,003	84.5	25.8	23.7–27.8
medical_debt	All respondents; current debt from medical care	12,934	0.0	18.3	17.6–19.0
unexpected_medical_expense	All respondents; unexpected major medical expense	12,934	0.0	21.3	20.5–22.0
low_buffer_both	Author construction: no \$400 cash/equivalent and no three-month savings	12,932	0.0	30.4	29.6–31.3

EF2 deserves special caution. It is not a population-wide rate. It is a follow-up item among adults without three-month rainy-day funds. The official SHED report presents the related concept in population terms: some adults without three-month funds could cover three months by borrowing, selling assets, or drawing on other savings, while others could not. This draft therefore does not use EF2 as a headline population estimate.

The paper also distinguishes “would pay” from “could pay.” The \$400 cash/equivalent measure captures how the adult says they would pay an emergency expense in the current situation. It is not a bank-balance audit. Some adults may have savings but prefer not to use them; others may use a credit card paid in full. This is a measurement feature, not a defect, but it limits interpretation.

5. Empirical strategy

The empirical strategy has three layers.

First, the paper reports weighted descriptive rates and approximate confidence intervals. The intervals use final weights as analysis weights and a Kish-style effective sample-size approximation. They are not replicate-weight

or full design-based intervals. SHED public-use files do not provide the same design information that would be needed for a fully design-based variance estimate in this simple Phase A build.

Second, the paper reports weighted cross-tabs by buffer status. These tables preserve unweighted cell sizes and highlight whether a rate is conditional.

Third, the paper adds approximate weighted linear-probability models. These models regress hardship outcomes on `low_buffer_both` with controls for income, education, age, race/ethnicity, housing tenure, employment, children, and region. The reported coefficients are average conditional associations in percentage points, not causal effects. Robust standard errors are approximate and should be read as a diagnostic improvement over point estimates alone, not as final survey-design inference.

6. Results

6.1 Weighted summary rates

Table 1. Weighted SHED 2025 summary rates.

Measure	Weighted rate (%)	Approx. 95% CI (%)	Interpretation
Cash/equivalent \$400 capacity	63.1	62.3–64.0	Would cover \$400 emergency with cash or equivalent.
Three-month emergency savings	54.8	53.9–55.7	Has rainy-day funds covering three months.
Credit card, paid over time	15.3	14.6–16.0	Would carry \$400 expense forward.
Unable to pay \$400 right now	12.2	11.6–12.8	Concrete inability-to-pay option.
Doing at least okay financially	72.9	72.1–73.7	General well-being indicator.
Financially vulnerable by B2	27.1	26.3–27.9	Just getting by or finding it difficult.
Bill pressure	27.8	27.0–28.6	Did not pay all bills in full or had difficult bills.
Outside-household help with expenses	23.2	22.4–24.0	Help with listed expenses.
Credit constrained	18.6	17.9–19.4	Desired credit but did not apply, denied, or limited.
Medical debt	18.3	17.6–19.0	Currently has debt from medical care.
Unexpected major medical expense	21.3	20.5–22.0	Unexpected out-of-pocket medical expense.
Lacks both core buffers	30.4	29.6–31.3	No \$400 cash/equivalent capacity and no three-month savings.

These rates broadly align with the official 2025 SHED report. The revised table adds two improvements: approximate uncertainty and concrete hardship outcomes beyond the general doing-okay measure.

6.2 Buffer status and hardship outcomes

Table 2. Selected weighted outcomes by emergency-buffer status.

Group	Doing okay (%)	Vulnerable (%)	Unable to pay \$400 (%)	Bill pressure (%)	Outside help (%)	Credit constrained (%)	Medical debt (%)
No \$400 cash/equivalent capacity	42.8	57.2	33.2	57.5	39.1	35.7	32.1
Has \$400 cash/equivalent capacity	90.5	9.5	0.0	10.5	13.9	8.7	10.2
No three-month emergency savings	50.4	49.6	25.1	48.8	33.9	31.3	27.7
Has three-month emergency savings	91.5	8.5	1.6	10.4	14.3	8.2	10.5
Does not lack both core buffers	88.6	11.4	1.3	13.0	15.8	9.9	12.0
Lacks both core buffers	37.1	62.9	37.2	61.5	40.1	38.6	32.8

The gaps are large across both subjective and concrete outcomes. Adults lacking both core buffers report financial vulnerability at 62.9 percent, bill pressure at 61.5 percent, credit constraints at 38.6 percent, and medical debt at 32.8 percent. Adults who do not lack both core buffers are much less exposed on each measure.

The inability-to-pay outcome is partly mechanically connected to the \$400 measure, so it should not be treated as independent validation. But bill pressure, outside help, credit constraints, and medical debt are less tautological. They show that the low-buffer group differs across several distinct hardship margins.

6.3 Approximate adjusted associations

Weighted cross-tabs can be confounded by income, tenure, age, education, and other respondent characteristics. Table 3 therefore reports approximate weighted linear-probability models. Each model includes `low_buffer_both` and controls for income, education, age, race/ethnicity, housing tenure, employment, children under 18, and region. Coefficients are percentage-point associations, not causal effects.

Table 3. Low-buffer association after controls.

Outcome	N	Low-buffer coefficient (pp)	Approx. robust SE (pp)	Approx. 95% CI (pp)
Financially vulnerable	12,932	38.3	1.1	36.1–40.6
Unable to pay \$400 right now	12,932	27.2	0.8	25.5–28.9
Bill pressure	12,932	36.1	1.1	33.9–38.4
Outside-household help	12,932	11.6	1.1	9.4–13.7
Credit constrained	12,932	21.8	1.1	19.7–23.9
Medical debt	12,932	19.0	1.1	16.9–21.1

The adjusted results reduce the risk that Table 2 is merely an income table in disguise. Even after standard demographic and household controls, lacking both core buffers is associated with substantially higher reported

hardship. The largest adjusted associations are for financial vulnerability and bill pressure. The association with outside help is smaller but still meaningful.

These adjusted estimates still do not identify causal effects. The unobserved factors named by the external review remain plausible: income volatility, health shocks, debt burdens, family networks, planning ability, financial literacy, risk preferences, and local cost conditions may shape both buffers and hardship.

6.4 BNPL as an auxiliary outcome only

The earlier version overused Buy Now, Pay Later late/fee outcomes as a confirming hardship proxy. The revised analysis moves BNPL out of the main argument.

The reason is selection. BNPL late/fee outcomes are observed among BNPL users with follow-up responses, not among all adults. The pattern is also not monotone across every buffer comparison. This does not make the variable useless, but it does make it unsuitable as a headline hardship indicator in a descriptive paper. The main results should rest on broader outcomes: bill pressure, inability to pay the \$400 expense, outside help, credit constraints, and medical debt.

7. Interpretation

The revised evidence supports a narrower claim than the original draft. Emergency-buffer measures are strongly associated with multiple hardship proxies in the 2025 SHED microdata. The association is visible in raw weighted cross-tabs and persists in approximate adjusted models.

The evidence does not show that buffers cause lower hardship. Reverse causality and omitted variables remain central concerns. A household may lack buffers because it has already experienced hardship. A health shock may simultaneously reduce savings and create bill pressure. Income volatility may lead to both weak buffers and credit constraints. Family support may reduce observed hardship even when liquid savings are low.

The descriptive result is still useful. If a simple low-buffer definition identifies a group with much higher bill pressure, credit constraint, outside help, and medical debt, then emergency-buffer measures are valuable for mapping household vulnerability. They are not the whole model of resilience, but they are not empty survey artifacts either.

8. Limitations

First, the paper is cross-sectional and descriptive. It cannot distinguish causality, reverse causality, or omitted-variable channels.

Second, some explanatory and outcome variables are conceptually close. “Doing at least okay financially” is a broad subjective self-assessment. It may partly incorporate perceived emergency capacity. The revised draft therefore emphasizes concrete hardship proxies, but even those remain self-reported.

Third, the \$400 measure is a “would pay” measure, not a “could pay from bank balances” measure. It includes credit-card use paid in full and can differ from savings-only capacity.

Fourth, skip patterns matter. EF2 and BNPL late/fee outcomes are conditional variables, not population-wide rates. They should be read with denominator information.

Fifth, the uncertainty estimates are approximate. They use final weights but not a full replicate-weight or design-based variance estimator.

Sixth, the paper uses 2025 only. It cannot assess trends, pandemic/post-pandemic changes, inflation-era changes, or credit-cycle dynamics.

9. Next empirical steps

The external review points to the correct path. The next version should not simply polish this 2025 note; it should extend it.

1. **Build a 2013–2025 SHED harmonized file.** This would allow annual trends and post-2021 comparisons. 2. **Prioritize hard outcomes.** Missed bills, inability to pay, medical hardship, food assistance, housing payment stress, and credit denial should matter more than general subjective well-being. 3. **Improve uncertainty.** If design variables or replicate-weight options are available, use them. Otherwise, clearly label bootstrap or robust approximations. 4. **Add robustness checks.** Separate checking/savings/cash from credit-card-paid-in-full; exclude card-paid-in-full from cash-equivalent capacity; estimate renter-only and owner-only models; stratify by income. 5. **Create a full replication package.** The package should include data provenance, software versions, one-command reproduction, a README mapping scripts to tables, and clean public outputs.

10. Conclusion

The revised Paper XI should be read as a descriptive microdata note, not a top-journal causal paper. Its value is that it converts a public-indicator emergency-savings discussion into transparent respondent-level evidence.

The main finding is robust descriptively: adults lacking both \$400 cash/equivalent capacity and three-month emergency savings report much higher hardship across several measures. They are more likely to be financially vulnerable, unable to pay a \$400 emergency right now, under bill pressure, receiving outside help with expenses, credit constrained, and carrying medical debt.

That pattern is not surprising, and it is not causal. But it is important. It shows that liquid emergency-buffer measures identify a population with a distinct hardship profile. The next research step is multi-year SHED harmonization and stronger modeling, not another broad conceptual paper.

Data and replication note

The analysis uses Federal Reserve SHED 2025 public-use CSV microdata and codebook. The replication workflow downloads the public ZIP, extracts selected variables, constructs derived indicators, and writes processed data and tables.

Main local outputs:

- `data/processed/shed_2025_selected_variables.csv`
- `data/processed/shed_2025_phase_a_manifest.json`
- `tables/shed_2025_variable_audit.csv`
- `tables/shed_2025_weighted_summary.csv`

- `tables/shed_2025_weighted_crosstabs.csv`
- `tables/shed_2025_weighted_subgroups.csv`
- `tables/shed_2025_adjusted_lpm.csv`
- `tables/shed_2025_phase_a_tables.md`

A public replication package should replace local-path references with repository-relative paths, include software versions, and provide one-command reproduction instructions.

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